**County matching: Glaeser and Gottlieb (2009)**

*Control: Match of Historical to Modern Geographic Units — Glaeser and Gottlieb (2009). At State level, this is a historical-instrument choice.*

**1. Match county identities.** This option matches historical county identities to current county units, using county codes and explicit identity corrections. Counties then enter their current MSA or, in State analysis, the historical FE group assigned in the dataset for the territory year. The regional membership is fixed when the instrument year changes. This is the project's Glaeser–Gottlieb-style county matching.

**2. Add whole historical populations.** Let $P_{jt}$ be the population of historical county $j$ in census year $t$, and let $m_{jg,t} = 1$ when that county is assigned to target region $g$. Otherwise, $m_{jg,t} = 0$. The matched population is

$$P_{g,t}^{\text{match}} = \sum_j m_{jg,t} P_{jt}.$$

Under Levels, the instrument is $z_{g,t} = P_{g,t}^{\text{match}}/1{,}000$. It measures thousands of people. Natural log instead uses $\log P_{g,t}^{\text{match}}$ for positive totals. The population total is used with both the employment and employment-density approaches. The minimum-overlap control does not apply to this method.

**3. Matching does not divide population by shared area.** If a historical county later splits, the match can assign its whole population to one identified successor. It does not distribute that population among all modern descendants. If all descendants remain in the same MSA, the MSA sum can nevertheless be unchanged. If a split crosses the modern MSA's boundary, matching and area allocation can produce different totals. The county identities, rather than the proportion of land inside the modern region, determine the assignment.

**4. Unresolved sources remain a limitation.** An uncertain identity is not assigned arbitrarily. Missing populations and known source gaps make affected totals unavailable. Jointly reported populations require special handling: matching names does not resolve an ambiguous allocation. A verified joint total is usable for a region only when its constituent units can all be assigned there and the total is counted once. Available totals and samples can therefore differ from area weighting. Compare methods on the same sample to separate population assignment from sample changes.